Exercise No. 5
CLASSICAL CIPHERS

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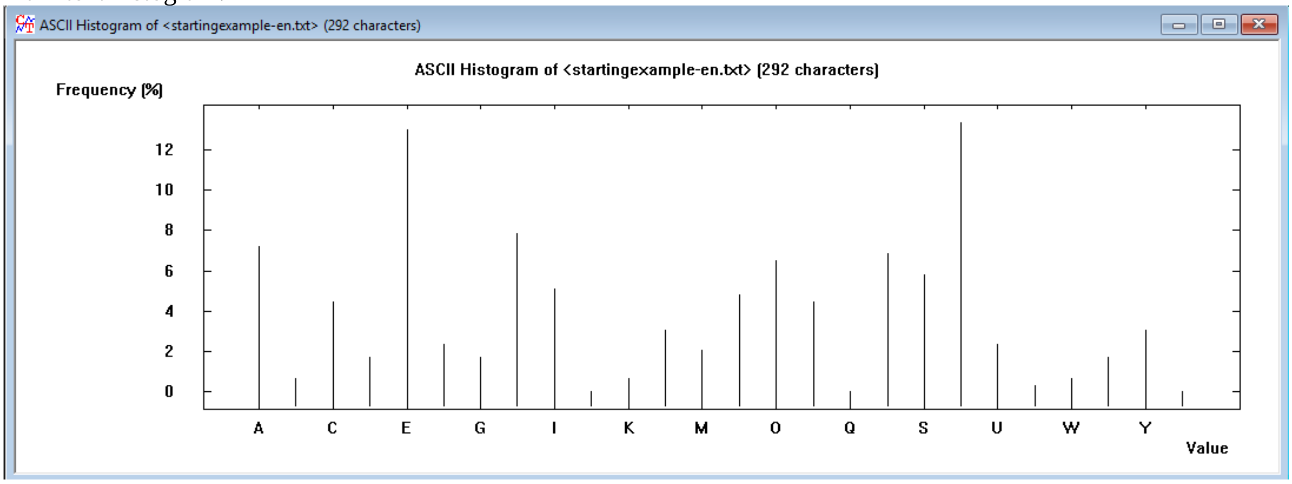
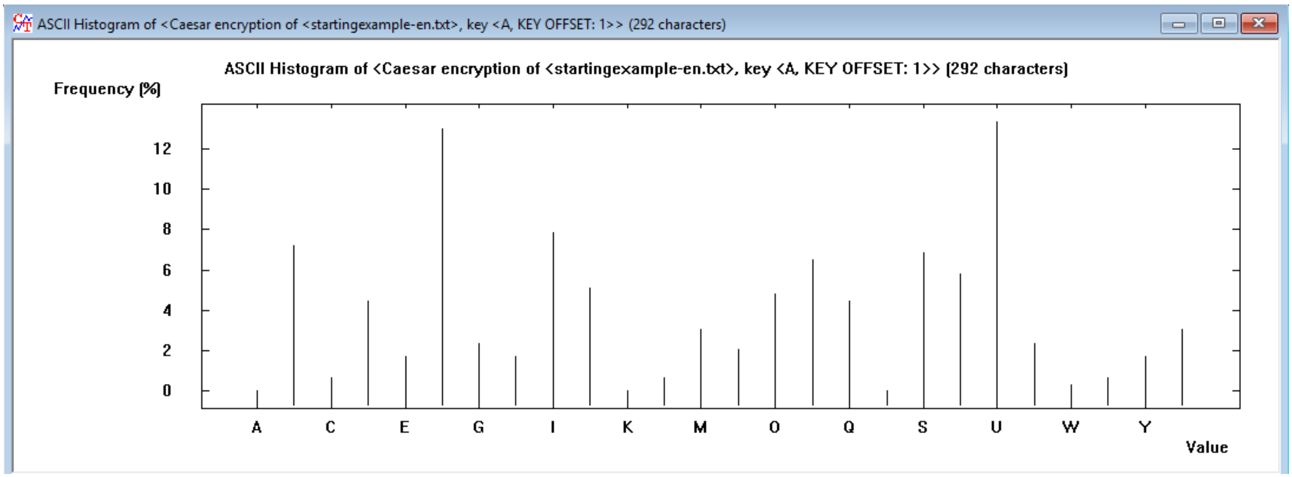
Task 3.1 Caesar Cipher

3.1.1

Plain text histogram:



Cyphered histogram:

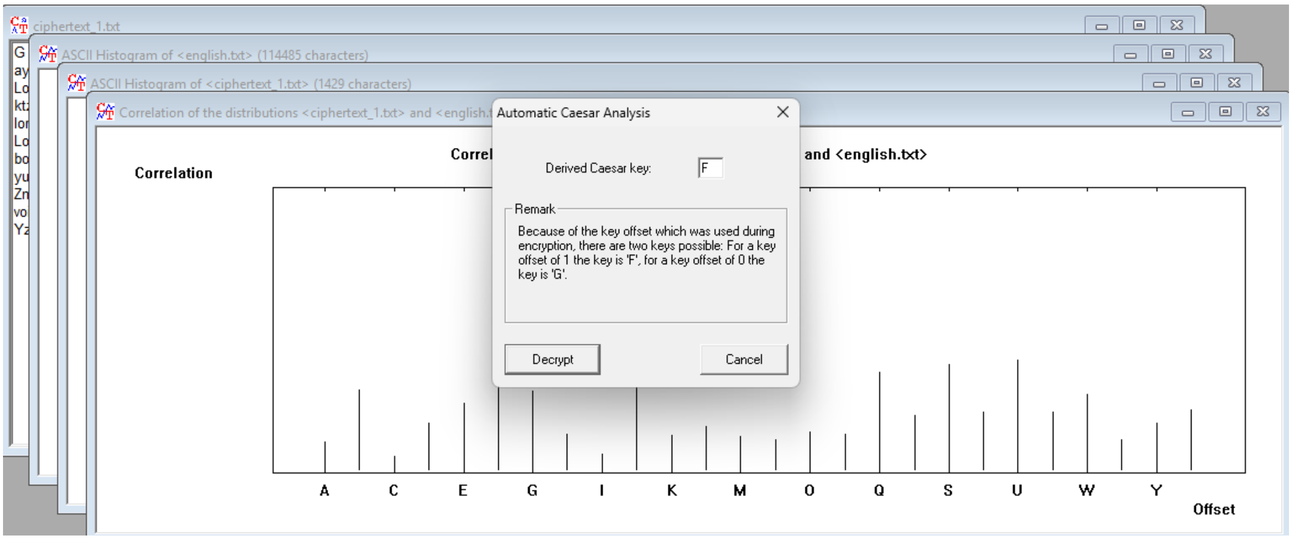


Conclusions:
- Caesar cypher is just shifting all characters by a same value so the histogram is almost identical.

3.1.2

In 26 ways so the key space is very low, this cypher could be brute-forced even by hand.

3.1.3



Cryptanalysis attack is comparing the histogram of Caesar cyphered text with the histogram of English words, and with this rule it derives the shift value that is most likely.

The cyphered text was about the: Filming industry.

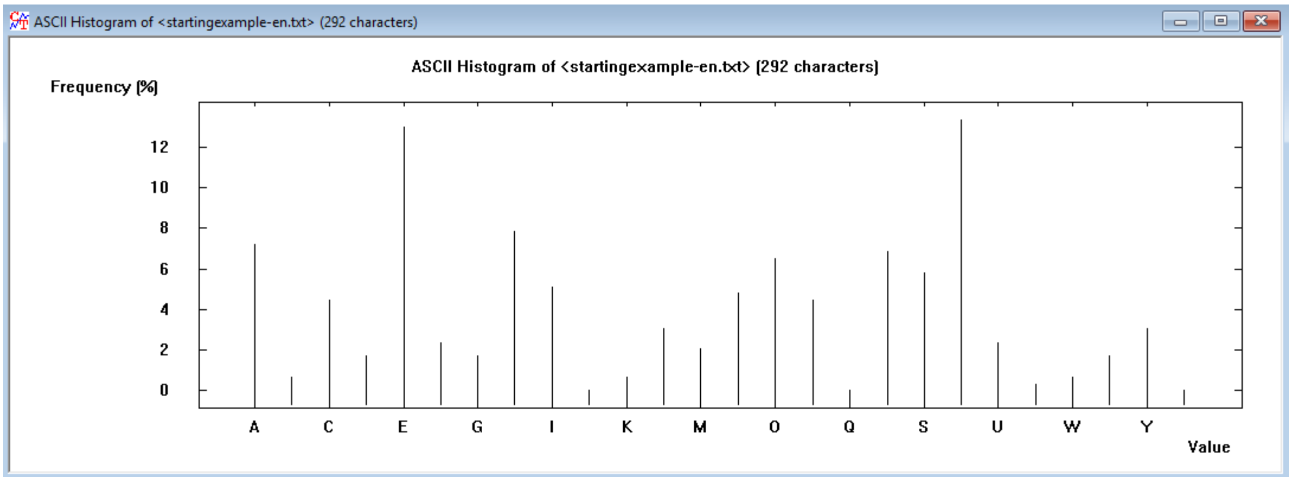
3.1.4
Advantages:
- It is very simple, so even a kid can perform it by hand – any cypher is better than no cypher at all

Disadvantages:
- It is very simple … So breaking one nowadays is no problem at all.

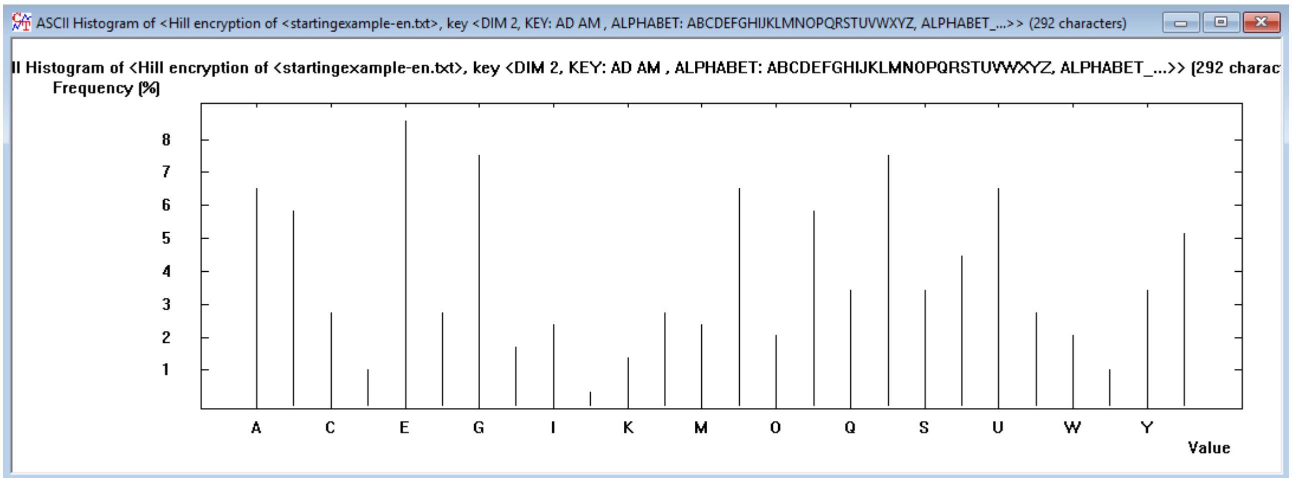
3.1.5
Rot13 is just a version of Caesar cypher, where shift key is set to 13.

3.2 Hill Cipher

3.2.1
Plain text:



Cyphered:

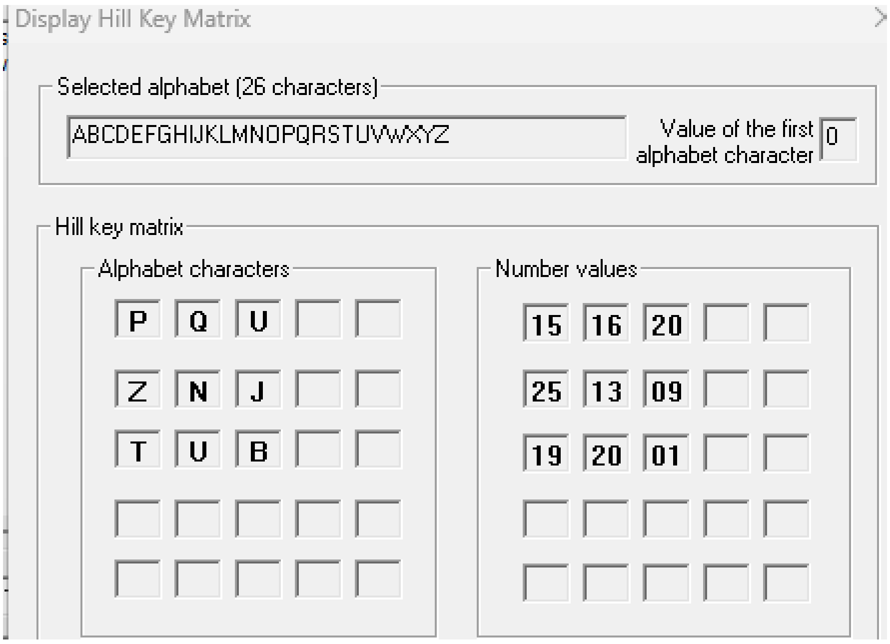


This time the cyphered histogram look much better than for Caesar cypher, the distribution at least for an eye seems to have some randomness included.

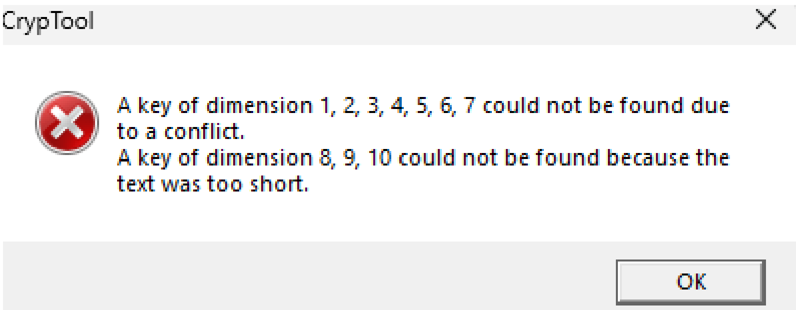
3.2.2
Not any, the requirement is coprimeness of the matrix determinant to 26 (e.g., 3,5,7,9,11,15)

3.2.3

Plaintext attack for files _1.



Plaintext attack for files _2.



Second attack was impossible due to conflicts and the text being to short. The text must contain n^2 letters where n is the dimension of the key matrix used.

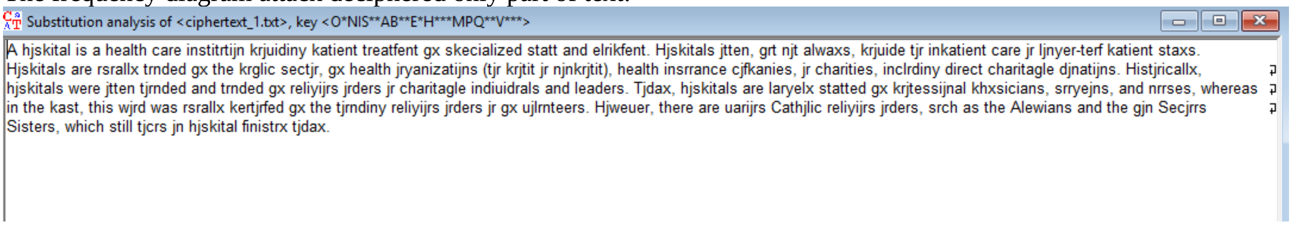
3.2.4
Advantages:
- Resists frequency analysis
- Is very fast

Disadvantages:
- Is vulnerable to known-plaintext-attack)
- Has tricky key selection
- Very weak nowadays

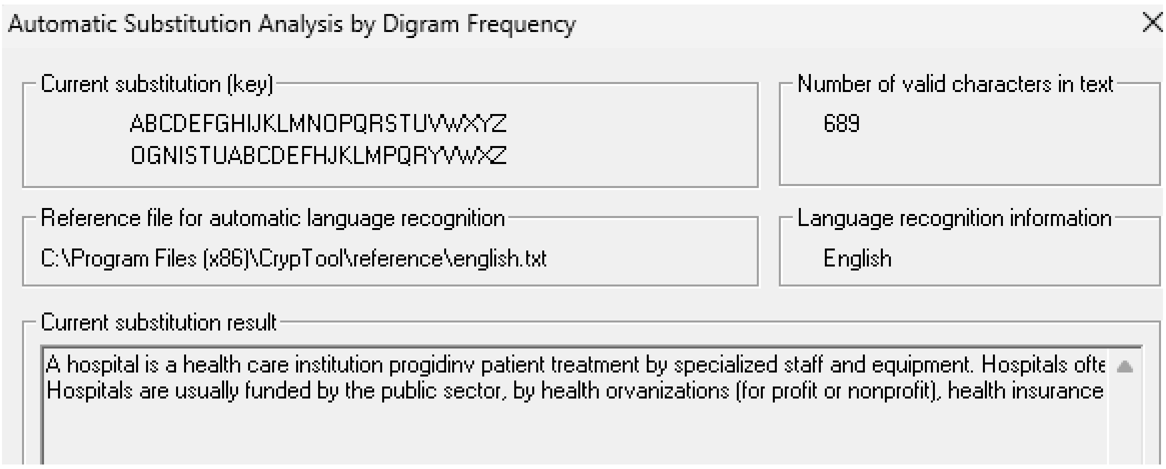
3.3 Monoalphabetic Substitution Cipher

3.3.1
You can encrypt a message in 26! ways: choosing one letter of 26 alphabet then one from 25 and so on…

3.3.2
The frequency diagram attack deciphered only part of text:



Most frequent words attack on the other hand deciphered whole text perfectly:



The text was about health and hospitals, and the key is visible on the second screenshot.

3.3.3
Advantages:
- huge key space
- easy implementation

Disadvantages:
- vulnerable to frequency attacks

3.3.4
General substitution allows for any random scrambling of the alphabet while the AtBash cipher is one specific fixed rule: reverse of the alphabet.

So the comparison would be like this:
Caesar Cipher: shifting an alphabet by a chosen number k modulo alphabet size
Rot 13: shifting alphabet always by 13 (for alphabet of 26 it is symmetrical so you can decrypt the text by encrypting it again)
AtBash: reversing the alphabet (again symmetrical – decrypting by encrypting again)

General Substitution: Arbitrary scrambling of the alphabet, far the most versatile of those 4 cyphers

3.4 Vigenére Cipher

3.4.1

Plaintext:

startingexample-en.txt

Using CrypTool, encrypt any plaintext (approx. 300 characters). As the encryption
key, use the first letter of your name. The value of the first character of the alphabet
should be set to 1 (A = 1). Then, decrypt the ciphertext using the same key. Display the
histogram of the plaintext and the histogram of the ciphertext. Compare the histograms.
Write down the conclusions.

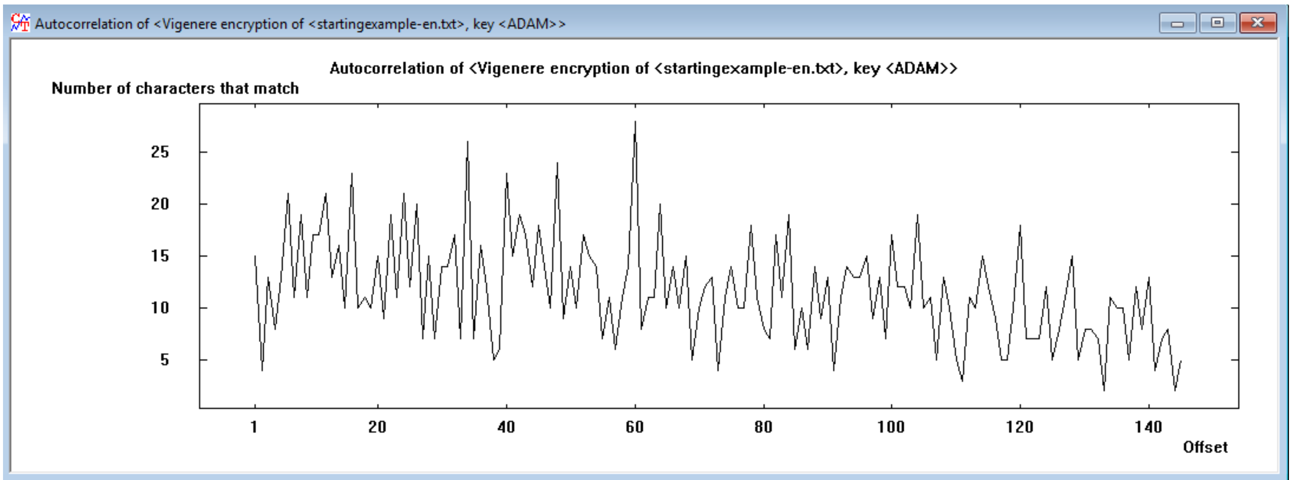
Cyphered:

Vigenere encryption of <startingexample-en.txt>, key <ADAM>

Uvizg FrkpWoal, hnorbpf aqy bldizthxf (aspdoa. 300 ctauaothre). Av tte hnorbpfirn
web, uee whq flret oefthr af bogr qaye. Whq vdlge rf fhh furvt ohdrmcwed oi tte dlbhdbqt
vhauod ne vef tr 1 (A = 1). Fhhn, pefrkpw tte fibhhrfeat gslns tke eape web. Dusslmy whq
hlsfojrmm rf fhh pxalnfeat mng tte kietrgdap or tke oishqrwejt. Foypdrq tke tivtaguays.
Zruth dawq tte fozcoueirne.

3.4.2

Autocorrelation:



The distance between peaks is equal to 4 - same as used key length. To break this cypher we could split the letters in 4 (distance) different columns and then perform a frequency analysis as those groups are basically a Caesar cyphers.